Amazon Worldwide Sustainability Data Dictionary

sanitized_2022-2025.csv

Dataset Overview

All datasets contain 47 columns with consistent schema across years.

Year	Row Count
2022	10,902,782
2023	11,557,004
2024	17,414,249
2025	18,981,023

Geographic / Location

Column Descriptions

Column	Description
country	Country code (US, CA, MX) – FC (Fulfillment Center) location
marketplace_id	Marketplace identifier - matches country values
hashed_fc	Hashed fulfillment center ID (anonymized)
country_state	State/province abbreviation where FC is located
zip_code	ZIP/postal code of FC location
fc_country	Unused placeholder column

Unique Values by Year

Column	2022	2023	2024	2025
country	3	3	3	3
marketplace_id	3	3	3	3
hashed_fc	unique	unique	unique	unique
fc_country	0	0	0	0
country_state	48	49	51	52
zip_code	~560	~560	~560	562

Missing Data % by Year

Column	2022	2023	2024	2025
country	0.00%	0.00%	0.00%	0.00%
marketplace_id	0.00%	0.00%	0.00%	0.00%
country_state	7.91%	9.00%	7.04%	7.52%
zip_code	7.94%	9.03%	7.06%	7.53%
fc_country	100%	100%	100%	100%

Time Dimensions

Column Descriptions

Column	Description
`year`	Year of the record
`month`	Month number (1-12)
`week`	Week number of the year (1-52)
`week_dt`	Week date placeholder (unused)

Unique Values by Year

Column	2022	2023	2024	2025
`year`	1	1	1	1
`month`	12	12	12	12
`week`	52	52	52	52
`week_dt`	0	0	0	0

Missing Data % by Year

Column	2022	2023	2024	2025
`year`	0.00%	0.00%	0.00%	0.00%
`month`	0.00%	0.00%	0.00%	0.00%
`week`	5.51%	6.22%	4.64%	4.37%
`week_dt`	100%	100%	100%	100%

Product Classification

Column Descriptions

Column	Description
`product_type`	High-level category: Food, Non Food, Pet Food
`macro_category`	Business model: RETAIL (third-party) or FBA (Amazon)
`gl_product_group`	General Ledger product group number
`product_type_granular`	Detailed product type (unused)
`gl_product_group_desc`	GL product group description (unused)
`category_code`	Product category code (unused)
`subcategory_code`	Product subcategory code (unused)
`asin`	Amazon Standard ID Number (anonymized)

Unique Values by Year

Column	2022	2023	2024	2025
`product_type`	3	3	3	3
`macro_category`	2	2	2	2
`gl_product_group`	138	135	135	142
`product_type_granular`	0	0	0	0
`gl_product_group_desc`	0	0	0	0
`category_code`	0	0	0	0
`subcategory_code`	0	0	0	0

asin	0	0	0	0

Missing Data % by Year

Column	2022	2023	2024	2025
product_type	0.00%	0.00%	0.00%	0.00%
macro_category	0.00%	0.00%	0.00%	0.00%
gl_product_group	0.02%	0.00%	0.01%	0.00%
product_type_granular	100%	100%	100%	100%
gl_product_group_desc	100%	100%	100%	100%
category_code	100%	100%	100%	100%
subcategory_code	100%	100%	100%	100%
asin	100%	100%	100%	100%

Disposition & Recovery

Column Descriptions

Column	Description
`item_disposition_code`	Item status: Sellable, Unsellable, Total, Overstock
`recovery_type`	Processing method: Donations, Liquidations, Return to Vendor, etc.
`reason_code`	Single-letter code: A, C, D, E, O, V, W, or Recall
`reason_code_type`	Reason category: ALSI, LLSI, MCF (available 2024+)

Unique Values by Year

Column	2022	2023	2024	2025
`item_disposition_code`	6	6	6	6
`recovery_type`	11	11	11	11
`reason_code`	8	8	8	8
`reason_code_type`	0	0	4	4

Missing Data % by Year

Column	2022	2023	2024	2025
`item_disposition_code`	15.18%	15.64%	38.51%	42.70%
`recovery_type`	0.00%	0.00%	0.00%	0.00%
`reason_code`	84.06%	84.35%	89.48%	87.84%
`reason_code_type`	100%	100%	71.47%	68.81%

Site / Facility Information

Column Descriptions

Column	Description
`site_type`	Facility type: Sort Center, Traditional FC, XL FC, AR Sortable, etc.
`site_category`	Site category: GCF, AMZL, AMXL, Fresh, GSF, Other

Unique Values by Year

Column	2022	2023	2024	2025
`site_type`	23	21	23	22
`site_category`	5	5	6	6

Missing Data % by Year

Column	2022	2023	2024	2025
`site_type`	7.91%	9.00%	7.04%	7.52%
`site_category`	7.91%	9.00%	7.04%	7.52%

Flags & Indicators

Column Descriptions

Column	Description

`is_stranded`	Y/N - Inventory stuck/cannot be processed (expanded 2024+)
`stranded_potential_issue`	Y/N - Potential issue with stranded inventory
`is_hazmat`	Y/N - Hazardous material indicator
`is_exception_destroy`	Exception destruction flag (unused)
`in_scope_ban`	Ban/restriction scope flag (unused)
`excl_from_reporting`	Exclude from reporting flag (unused)
`is_pod`	Print on Demand flag (unused)
`is_expiration_dated_product`	Expiration dated product flag (unused)

Unique Values by Year

Column	2022	2023	2024	2025
`is_stranded`	1	1	2	2
`stranded_potential_issue`	1	1	2	2
`is_hazmat`	2	2	2	2
`is_exception_destroy`	0	0	0	0
`in_scope_ban`	0	0	0	0
`excl_from_reporting`	0	0	0	0
`is_pod`	0	0	0	0
`is_expiration_dated_product`	0	0	0	0

Missing Data % by Year

Column	2022	2023	2024	2025
`is_stranded`	84.82%	84.36%	61.49%	57.30%
`stranded_potential_issue`	84.82%	84.36%	61.49%	57.30%
`is_hazmat`	0.00%	0.00%	0.00%	0.00%
`is_exception_destroy`	100%	100%	100%	100%
`in_scope_ban`	100%	100%	100%	100%
`excl_from_reporting`	100%	100%	100%	100%
`is_pod`	100%	100%	100%	100%
`is_expiration_dated_product`	100%	100%	100%	100%

Stranded Inventory Details

Column Descriptions

Column	Description
`reason_code_stranded`	Detailed stranded reason codes (available 2024+)
`stranded_sub_reason`	Sub-reason for stranded inventory (unused)

Unique Values by Year

Column	2022	2023	2024	2025
`reason_code_stranded`	0	0	82	78
`stranded_sub_reason`	0	0	0	0

Missing Data % by Year

Column	2022	2023	2024	2025

reason_code_stranded	100%	100%	71.47%	68.81%
stranded_sub_reason	100%	100%	100%	100%

System & Vendor Information

Column Descriptions

Column	Description
application_name	Processing system: DeleteItemsApp, WAVE_DPT (2023+), MurfiV2 (2024+)
source_id	Source identifier (unused)
vendor_code	Vendor identifier code (anonymized)
vendor_name	Vendor name (anonymized)

Unique Values by Year

Column	2022	2023	2024	2025
application_name	1	2	3	3
source_id	0	0	0	0
vendor_code	0	0	0	0
vendor_name	0	0	0	0

Missing Data % by Year

Column	2022	2023	2024	2025
application_name	87.85%	91.32%	92.36%	97.22%
source_id	100%	100%	100%	100%
vendor_code	100%	100%	100%	100%
vendor_name	100%	100%	100%	100%

Metrics / Measurements

Column Descriptions

Column	Description
units	Number of units/quantity
cogs	Cost of Goods Sold (dollar value)
weight	Weight of items (likely lbs or kg)

Unique Values by Year

Column	2022	2023	2024	2025
units	~689K	~700K	~750K	~689K
cogs	~1.4M	~1.5M	~1.6M	~1.4M
weight	~7.9M	~8M	~8.2M	~8M

Missing Data % by Year

Column	2022	2023	2024	2025
units	0.00%	0.00%	0.00%	0.00%
cogs	0.00%	0.00%	0.00%	0.00%

weight	0.07%	0.02%	0.04%	0.03%

Appendix: Categorical Value Reference

country / marketplace_id

Value	Present In
CA	2022, 2023, 2024, 2025
MX	2022, 2023, 2024, 2025
US	2022, 2023, 2024, 2025

product_type

Value	Present In
Food	2022, 2023, 2024, 2025
Non Food	2022, 2023, 2024, 2025
Pet Food	2022, 2023, 2024, 2025

macro_category

Value	Present In
FBA	2022, 2023, 2024, 2025
RETAIL	2022, 2023, 2024, 2025

item_disposition_code

Value	Present In
Sellable	2022, 2023, 2024, 2025
SELLABLE	2022, 2023, 2024, 2025
Unsellable	2022, 2023, 2024, 2025
UNSELLABLE	2022, 2023, 2024, 2025
Overstock	2022, 2023, 2024, 2025
Total	2022, 2023, 2024, 2025

recovery_type

Value	Present In
Bintool Donations	2022, 2023, 2024, 2025
Bintool Remove Liquidate	2022, 2023, 2024, 2025
Bintool Theft	2022, 2023, 2024, 2025
C-Returns	2022, 2023, 2024, 2025
Donations	2022, 2023, 2024, 2025
Liquidations	2022, 2023, 2024, 2025
Remove Liquidate	2022, 2023, 2024, 2025
Remove Return	2022, 2023, 2024, 2025
Return to Vendor	2022, 2023, 2024, 2025
Sales	2022, 2023, 2024, 2025
Warehouse Deals and G&R	2022, 2023, 2024, 2025

reason_code

Value	Present In

A	2022, 2023, 2024, 2025
C	2022, 2023, 2024, 2025
D	2022, 2023, 2024, 2025
E	2022, 2023, 2024, 2025
O	2022, 2023, 2024, 2025
V	2022, 2023, 2024, 2025
W	2022, 2023, 2024, 2025
Recall	2022, 2023, 2024, 2025

site_category

Value	Present In
AMXL	2022, 2023, 2024, 2025
AMZL	2022, 2023, 2024, 2025
Fresh	2022, 2023, 2024, 2025
GCF	2022, 2023, 2024, 2025
GSF	2024, 2025 only
Other	2022, 2023, 2024, 2025

application_name

Value	Present In
DeleteItemsApp	2022, 2023, 2024, 2025
WAVE_DPT	2023, 2024, 2025 only
MurfiV2	2024, 2025 only

reason_code_type

Value	Present In
ALSI	2024, 2025 only
LLSI	2024, 2025 only
MCF	2024, 2025 only
\N (null)	2024, 2025 only

Open Questions

1. What do single-letter reason codes mean? (A, C, D, E, O, V, W)
2. What is the difference between ALSI, LLSI, MCF in reason_code_type?
3. What does "stranded" inventory mean in this context?
4. What do the values “New”, “Nor”, and “TB” represent in country_state?
5. What do the ~80 codes in reason_code_stranded mean?
6. What do the three applications in application_name represent?
7. What to do with the 100% empty columns? Are they supposed to be empty?
8. gl_product_group appears to be ~142 numeric codes for product types. Might need clarification on what they represent if we need to do material analysis for waste cost approximation.
9. Since recovery_type is likely our target variable, would be beneficial to completely clarify each value in that column. (refer to Appendix, recovery_type)
10. Which fields are determined before/after the target variable recovery_type is decided on. Ex. Is_stranded, reason_code_stranded